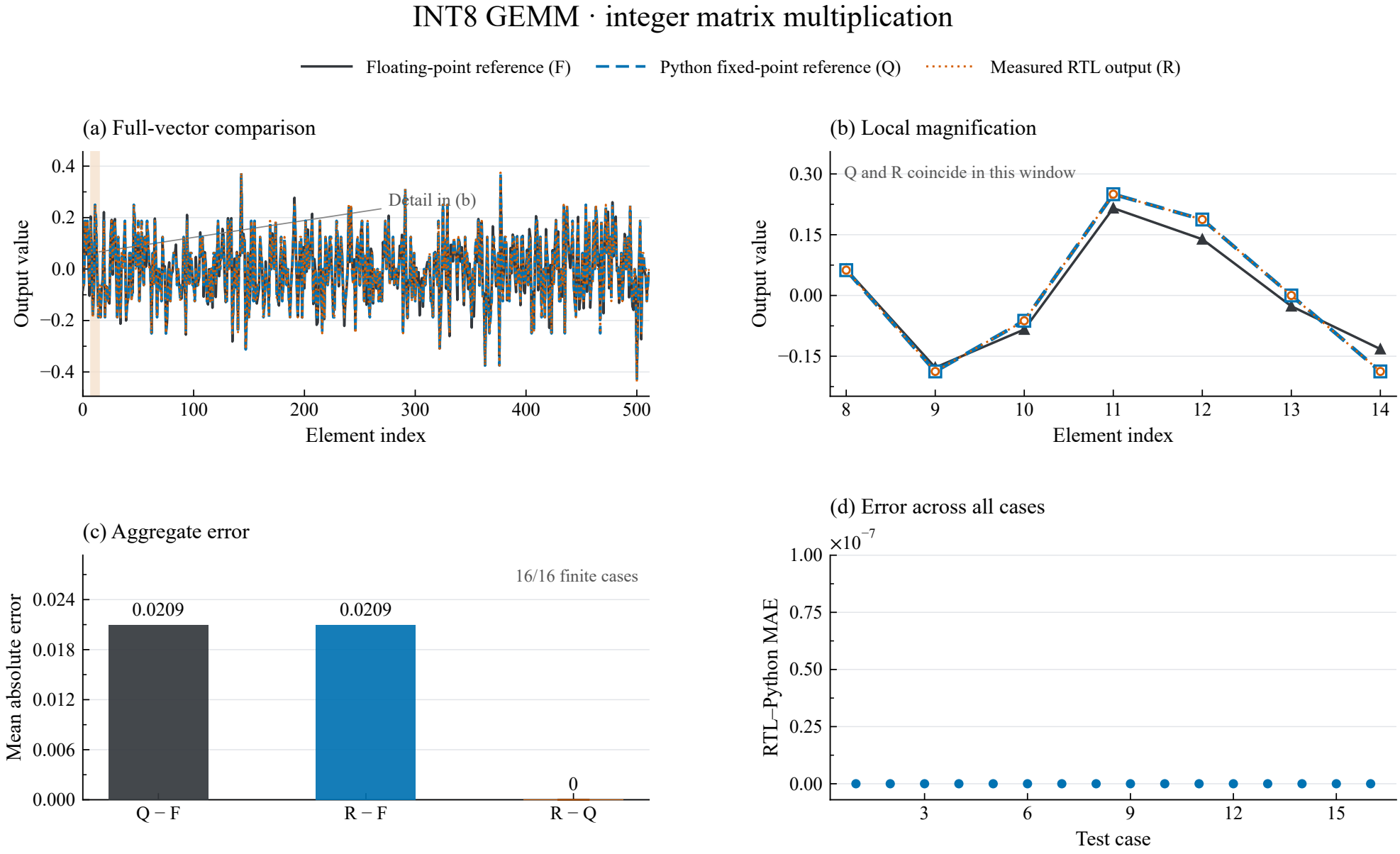
INT8 GEMM · integer matrix multiplication
Floating-point reference (F)
Python fixed-point reference (Q)
Measured RTL output (R)
(a) Full-vector comparison
Output value
Element index
Detail in (b)
(b) Local magnification
Q and R coincide in this window
Output value
Element index
(c) Aggregate error
16/16 finite cases
Mean absolute error
0.0209
0.0209
0
Q − F
R − F
R − Q
(d) Error across all cases
×10−7
RTL–Python MAE
Test case
16 cases · input Q4 / output Q4 · example #5: M=16 / K=32 / N=32 · cases 9–16 include stalls.
No vertical offsets. Red markers retain failed checks; all timing values are reported in tables.